Checkpoint Report – Neeraj Arde, Kriish Tiwari, Prasanna Govindarajulu Sairam

Design of a 16-bit Processor

https://github.com/neerajArde/cse490-16bit-processor.git

Week 1 Progress

Our project goal is to design and implement a 16-bit single-cycle processor using Verilog. The processor will execute instructions according to a defined instruction set architecture and will include all the required and necessary Datapath components. The processor will be expected to fetch, decode, execute, and finish instructions using a single clock cycle.

To get this finished, we looked through all the necessary architecture and talked about the hardware components needed for this specific processor design. These components include the Arithmetic Logic Unit (ALU), register file, program counter (PC), instruction memory, data memory, control unit, sign-extension unit, and multiplexers. Us understanding how these components come together as a whole in a datapath was a really important part of our planning phase at the beginning.

At the beginning of the project, our team created a shared GitHub repository to make sure all of our project materials were neat and together. This then allows collaboration between group members. The repository contains folders for source code, documentation, testbenches, and project progress files. This structure allows the team to organize files clearly and track progress throughout development.

The repository includes the following main folders:

- **WorkDivision** – documents describing task assignments and team responsibilities
- **WeeklyLog** – records of weekly meetings and development progress
- **SourceCode** – all Verilog modules and implementation files
- **ProjectReport** – checkpoint and final project reports

This repository setup ensures that all project work is all together and is accessible to team members and course staff.

Components Started:

So far, the team has started planning and preparing the processor implementation. We first found all the datapath components that is completely needed for the processor. This includes the ALU, register file, program counter, memory modules, and control logic. After finding out the needed components, we planned the structure of the Verilog modules that will be used to implement each part of the datapath. The team also talked about how instructions will flow through the processor during execution, including how they will be fetched, decoded, and processed within the datapath. We also talked about how the processor will be tested using Verilog testbenches and simulation tools before eventually deploying the design to hardware.

Work Division:

We decided to split up the work in a way that everyone gets to design each aspect of our overall design. Each member will work on specific modules of the processor, allowing components to be developed independently and later integrated into the full datapath.

The task distribution includes:

- ALU implementation
- Register file design
- Program counter logic
- Control unit development
- Memory modules
- Integration and testing

This division of work allows the project to progress while maintaining collaboration through the shared repository.

Our Next Steps:

The next phase of our project will focus on implementing individual datapath components.

Planned next steps include the following:

- Implementing the ALU module in Verilog
- Implementing the Register File
- Implementing the Program Counter
- Writing testbenches to verify each module
- Integrating modules into the full processor datapath

Once the datapath is integrated and tested in simulation, the design will eventually be deployed onto an FPGA board.